

JOHN PAUL S

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EDUCATION

B.Tech in Information Technology

Prince Shri Venkateshwara Padmavathy Engineering College (PSVPEC), Chennai

Affiliated to Anna University CGPA: 6.9/10

Chennai

TECHNICAL SKILLS

Programming

Python C++ Java

Deep Learning

PyTorch

Computer Vision

OpenCV

XR Development

Unity Unreal Engine

Systems

SDN Concepts Fog Computing Blockchain Basics

Mathematics

Spectral Analysis Optimization Constraints

RESEARCH INTERESTS

Physics-Guided Deep Learning | Underwater Image Restoration |
Computer Vision & Visual Geometry | Immersive Multimedia Systems (VR/AR) |
Fog-Blockchain Medical IoT Architectures | Uncertainty-Aware Neural Networks

RESEARCH WORK

APGR-Net: Adaptive Physics-Guided Restoration for Underwater Images

A novel deep learning framework for single-image underwater restoration under turbulent water distortion.

Core Innovations

Multi-Hypothesis Warping

- Predicts three warp fields (dx, dy)
- Confidence-weighted blending
- Handles ambiguous distortion fields

Physics-Informed Constraints

- Snell's law angle limit ($\leq 48.6^\circ$)
- Surface smoothness constraint
- Depth-dependent maximum displacement
- Penalized physically impossible warps

Distortion-Aware Attention

- Severity map guides adaptive feature allocation

Caustics-Aware Processing

- Detects and removes bright water-surface artifacts

Uncertainty Quantification

- Monte Carlo Dropout inference
- Pixel-level confidence estimation

Representation Stability Analysis

- Effective Rank = 92.84
- Spectral entropy evaluation
- Condition number = 65
- Energy ratio validation
- Non-collapsed embedding confirmation

Designed low-cost data protocol (~\$100 setup) generating ~810,000 training pairs. Manuscript under preparation.

SU-CARE-FoB: Secure Urgency-Aware Routing for Fog-Blockchain Medical IoT

A scalable healthcare architecture integrating Fog Intelligence, SDN routing, and Blockchain auditing.

Contributions

- PUF-based device authentication
- SHA-256 verification
- Fog-level severity scoring
- Sliding window smoothing
- Persistence modeling
- Network-aware evaluation
- SDN-based adaptive routing
- 48-hour summarized blockchain logging

Outcomes

- Reduced emergency response latency
- Lower false alarm rate
- Selective blockchain usage
- Congestion-aware routing
- Secure IoT communication

Architecture paper in structured development.

Review Paper (Accepted - International Conference)

Systematic Review of Secure and Scalable Fog-Blockchain Architectures in Medical IoT Systems

Covers:

- Fog scalability models
- Blockchain overhead mitigation
- Secure healthcare routing mechanisms
- Latency-aware emergency detection

Accepted for presentation at an international conference.

RESEARCH & INDUSTRIAL EXPERIENCE

AI Intern - CRION Technology (IIT Madras Associated Initiative)

Worked on AI-driven automation systems.

Contributions

- Food package type detection using vision models
- OCR-based text extraction pipeline
- Structured documentation system
- Modular Python-based model design
- Neural network model development and deployment

Research Intern - National Taiwan University (TEEP Program, RTRC Lab)

Developed a deep learning model for underwater image restoration.

- CNN-based geometric correction
- Color distortion compensation
- Turbidity-aware restoration
- Experimental evaluation under multiple water conditions

This work evolved into APGR-Net.

Intern - Trinitysoft Solutions Pvt Ltd (UK)

- Developed full-stack web application
- Deployed cloud-based LLM integration
- Backend API + frontend communication

SELECTED PROJECTS

Smart Drone Navigation System

AI-based drone routing system incorporating:

- Dynamic charge-aware navigation
- Pickup-drop optimization
- Charging station decision logic
- Route feasibility analysis

AI-Driven Plant Care System

- IoT sensor integration
- Soil moisture detection
- Automated irrigation logic
- AI-based plant health monitoring

LegaBud - AI Legal Document Processor

LLM-powered system for legal text analysis.

- Mistral-based processing
- Constraint extraction
- Full-stack API integration

AI Summarization Tool

- LORA fine-tuned Mistral model
- Word-constrained summarization
- Custom training dataset creation

Anonymous Real-Time Chat Platform

- Privacy-focused architecture
- Secure frontend-backend communication
- No persistent identity logging

RESEARCH ALIGNMENT

- My work combines physics-constrained visual modeling, uncertainty-aware learning, and scalable intelligent architectures. These directly support immersive multimedia systems where geometric correctness, robustness, and real-time performance are critical.
- I am particularly interested in extending physics-informed reconstruction methods into immersive AR/VR visual computing environments.